

CSS prefers-color-scheme Complete Guide

CSS Educational Series

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Chapter 1

Introduction to prefers-color-scheme

1.1 What is prefers-color-scheme?

The CSS `prefers-color-scheme` media query detects whether the user has requested a light or dark color scheme. This is based on the operating system or browser settings, allowing websites to adapt their appearance to match user preferences without requiring manual theme toggles.

1.1.1 Key Facts

- Detects OS/browser dark mode preference
- Values: `light`, `dark`, `no-preference`
- Respects system-wide accessibility settings
- No JavaScript required for basic implementation
- Improves user experience and accessibility
- Reduces eye strain for users preferring dark mode
- Modern browser support: All modern browsers (90
- Works with CSS custom properties for dynamic theming

1.1.2 Why prefers-color-scheme Matters

1. **Accessibility:** Respects user accessibility preferences
2. **User Comfort:** Reduces eye strain by matching preference
3. **Energy Efficiency:** Dark mode saves battery on OLED displays
4. **System Integration:** Works seamlessly with OS theme settings
5. **User Control:** Users manage preference at OS level
6. **Professional:** Expected in modern web applications
7. **Analytics:** Understand user preferences for design decisions

1.1.3 Browser Support

Browser	Chrome	Firefox	Safari	Edge
prefers-color-scheme	76+	67+	12.1+	79+

Chapter 2

Basic Syntax and Concepts

2.1 Fundamental Syntax

```
1  /* Light mode (default) */
2  @media (prefers-color-scheme: light) {
3      body {
4          background-color: white;
5          color: #333;
6      }
7  }
8
9  /* Dark mode */
10 @media (prefers-color-scheme: dark) {
11     body {
12         background-color: #1e1e1e;
13         color: #e0e0e0;
14     }
15 }
16
17 /* No preference (treat as light) */
18 @media (prefers-color-scheme: no-preference) {
19     body {
20         background-color: white;
21         color: #333;
22     }
23 }
24
25 /* Alternative: only dark mode */
26 @media (prefers-color-scheme: dark) {
27     * {
28         --bg: #000;
29         --text: #fff;
30     }
31 }
```

Listing 2.1: Basic prefers-color-scheme Usage

2.2 Media Query Syntax Details

```
1  /* Single scheme */
2  @media (prefers-color-scheme: dark) {
3      /* Rules for dark mode */
4  }
5
6  /* Multiple conditions */
7  @media (prefers-color-scheme: dark) and (max-width: 768px) {
8      /* Dark mode on mobile */
9  }
10
11 /* Negation */
12 @media not (prefers-color-scheme: dark) {
13     /* Everything except dark mode */
14 }
15
16 /* Using in HTML meta tags */
17 <!-- Suggest browser use light theme for manifest -->
18 <meta name="theme-color" content="white"
19       media="(prefers-color-scheme: light)">
20 <meta name="theme-color" content="#1e1e1e"
21       media="(prefers-color-scheme: dark)">
```

Listing 2.2: Media Query Structure

2.3 How It Works

```
1  /* Browser respects these system settings: */
2  /* Windows: Settings > Colors > Light/Dark */
3  /* macOS: System Preferences > General > Appearance */
4  /* iOS: Settings > Display & Brightness > Dark Mode */
5  /* Android: Settings > Display > Dark theme */
6  /* CSS detects this and applies matching styles */
7
8  @media (prefers-color-scheme: light) {
9      /* User has OS light mode OR browser light setting */
10     body { background: white; }
11 }
12
13 @media (prefers-color-scheme: dark) {
14     /* User has OS dark mode OR browser dark setting */
15     body { background: #1e1e1e; }
16 }
17
18 /* If system says "light" → light styles apply */
19 /* If system says "dark" → dark styles apply */
```

Listing 2.3: Understanding Preference Detection

Chapter 3

Color Systems and Theming

3.1 CSS Custom Properties for Theming

```
1 <style>
2   /* Light theme (default) */
3   :root {
4     --bg-primary: #ffffff;
5     --bg-secondary: #f5f5f5;
6     --text-primary: #1a1a1a;
7     --text-secondary: #666666;
8     --border-color: #e0e0e0;
9     --accent: #0066cc;
10    --success: #00a86b;
11    --error: #d32f2f;
12    --warning: #ff9800;
13  }
14
15  /* Dark theme */
16  @media (prefers-color-scheme: dark) {
17    :root {
18      --bg-primary: #1e1e1e;
19      --bg-secondary: #2d2d2d;
20      --text-primary: #e0e0e0;
21      --text-secondary: #b0b0b0;
22      --border-color: #404040;
23      --accent: #4da6ff;
24      --success: #66cc99;
25      --error: #ff6b6b;
26      --warning: #ffb74d;
27    }
28  }
29
30  body {
31    background-color: var(--bg-primary);
32    color: var(--text-primary);
33  }
34
35  .card {
36    background-color: var(--bg-secondary);
```



```

37     border: 1px solid var(--border-color);
38     color: var(--text-secondary);
39 }
40
41 .button {
42     background-color: var(--accent);
43     color: white;
44 }
45
46 .alert-success {
47     background-color: var(--success);
48     color: white;
49 }
50
51 .alert-error {
52     background-color: var(--error);
53     color: white;
54 }
55 </style>
56
57 <body>
58     <div class="card">
59         <h2>Themed Content</h2>
60         <p>This card uses CSS custom properties that adapt to system theme.</p>
61         <button class="button">Learn More</button>
62     </div>
63
64     <div class="alert-success">Success message</div>
65     <div class="alert-error">Error message</div>
66 </body>

```

Listing 3.1: Theme Variables with prefers-color-scheme

3.2 Comprehensive Color Palette

```

1 <style>
2 :root {
3     /* Light mode */
4     --color-surface: #ffffff;
5     --color-surface-variant: #f5f5f5;
6     --color-surface-dim: #efefef;
7
8     --color-on-surface: #1a1a1a;
9     --color-on-surface-variant: #666666;
10
11     --color-primary: #0066cc;
12     --color-primary-container: #e0efff;
13     --color-on-primary: #ffffff;
14
15     --color-secondary: #006194;
16     --color-secondary-container: #cce5ff;

```

```
17     --color-on-secondary: #ffffff;
18
19     --color-tertiary: #8d4e01;
20     --color-tertiary-container: #ffdcc9;
21     --color-on-tertiary: #ffffff;
22
23     --color-error: #d32f2f;
24     --color-error-container: #ffebee;
25     --color-on-error: #ffffff;
26
27     --color-outline: #ccc;
28     --color-outline-variant: #e0e0e0;
29 }
30
31 @media (prefers-color-scheme: dark) {
32     :root {
33         --color-surface: #1a1a1a;
34         --color-surface-variant: #2d2d2d;
35         --color-surface-dim: #0f0f0f;
36
37         --color-on-surface: #e0e0e0;
38         --color-on-surface-variant: #b0b0b0;
39
40         --color-primary: #4da6ff;
41         --color-primary-container: #005cc3;
42         --color-on-primary: #000000;
43
44         --color-secondary: #99cefd;
45         --color-secondary-container: #004472;
46         --color-on-secondary: #000000;
47
48         --color-tertiary: #ffc9a1;
49         --color-tertiary-container: #6b3800;
50         --color-on-tertiary: #000000;
51
52         --color-error: #ff6b6b;
53         --color-error-container: #8b0000;
54         --color-on-error: #ffffff;
55
56         --color-outline: #555;
57         --color-outline-variant: #404040;
58     }
59 }
60
61 body {
62     background: var(--color-surface);
63     color: var(--color-on-surface);
64 }
65
66 .button-primary {
67     background: var(--color-primary);
68     color: var(--color-on-primary);
69 }
70 </style>
```

Listing 3.2: Complete Color System

Chapter 4

Practical Applications

4.1 Complete Website Theme

```
1 <style>
2   * {
3     margin: 0;
4     padding: 0;
5     box-sizing: border-box;
6   }
7
8   :root {
9     --bg: #fff;
10    --bg-secondary: #f5f5f5;
11    --text: #1a1a1a;
12    --text-light: #666;
13    --border: #ddd;
14    --shadow: rgba(0,0,0,0.1);
15  }
16
17  @media (prefers-color-scheme: dark) {
18    :root {
19      --bg: #1a1a1a;
20      --bg-secondary: #2d2d2d;
21      --text: #e0e0e0;
22      --text-light: #999;
23      --border: #404040;
24      --shadow: rgba(0,0,0,0.3);
25    }
26  }
27
28  body {
29    background: var(--bg);
30    color: var(--text);
31    font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', sans-serif;
32    ;
33    transition: background 0.3s, color 0.3s;
34  }
35
36  header {
```

```
36     background: var(--bg-secondary);
37     border-bottom: 1px solid var(--border);
38     padding: 20px;
39     box-shadow: 0 1px 3px var(--shadow);
40 }
41
42 nav {
43     display: flex;
44     gap: 20px;
45 }
46
47 nav a {
48     color: var(--text);
49     text-decoration: none;
50     font-weight: 500;
51     transition: color 0.2s;
52 }
53
54 nav a:hover {
55     color: #0066cc;
56 }
57
58 main {
59     max-width: 1200px;
60     margin: 40px auto;
61     padding: 0 20px;
62 }
63
64 .card {
65     background: var(--bg-secondary);
66     border: 1px solid var(--border);
67     border-radius: 8px;
68     padding: 20px;
69     margin-bottom: 20px;
70     box-shadow: 0 2px 8px var(--shadow);
71 }
72
73 .card h3 {
74     color: var(--text);
75     margin-bottom: 10px;
76 }
77
78 .card p {
79     color: var(--text-light);
80     line-height: 1.6;
81 }
82
83 button {
84     background: #0066cc;
85     color: white;
86     border: none;
87     padding: 10px 20px;
88     border-radius: 4px;
89     cursor: pointer;
```

```
90     font-weight: 600;
91     transition: background 0.2s;
92 }
93
94 button:hover {
95     background: #0052a3;
96 }
97
98 footer {
99     background: var(--bg-secondary);
100    border-top: 1px solid var(--border);
101    padding: 20px;
102    text-align: center;
103    color: var(--text-light);
104    margin-top: 40px;
105 }
106 </style>
107
108 <header>
109     <nav>
110         <a href="#home">Home</a>
111         <a href="#about">About</a>
112         <a href="#contact">Contact</a>
113     </nav>
114 </header>
115
116 <main>
117     <h1>Welcome to Themed Website</h1>
118     <p>This site automatically adapts to your system theme preference.</p>
119
120     <div class="card">
121         <h3>Light and Dark Modes</h3>
122         <p>Your system preference is automatically detected. Change your OS
123             theme settings and watch the site adapt in real-time.</p>
124         <button>Learn More</button>
125     </div>
126
127     <div class="card">
128         <h3>Consistent Design</h3>
129         <p>All elements maintain visual hierarchy and readability in both
130             light and dark modes.</p>
131     </div>
132 </main>
133
134 <footer>
135     <p>&copy; 2024 Themed Website. All rights reserved.</p>
136 </footer>
```

Listing 4.1: Full-Featured Website with Dark Mode

4.2 Blog Theme

```
1 <style>
2   :root {
3     --article-bg: #ffffff;
4     --article-text: #222;
5     --article-heading: #000;
6     --code-bg: #f5f5f5;
7     --code-text: #333;
8     --link-color: #0066cc;
9     --blockquote-border: #ddd;
10  }
11
12  @media (prefers-color-scheme: dark) {
13    :root {
14      --article-bg: #1e1e1e;
15      --article-text: #d0d0d0;
16      --article-heading: #ffffff;
17      --code-bg: #2d2d2d;
18      --code-text: #e0e0e0;
19      --link-color: #4da6ff;
20      --blockquote-border: #555;
21    }
22  }
23
24  article {
25    background: var(--article-bg);
26    color: var(--article-text);
27    padding: 40px;
28    max-width: 800px;
29    margin: 0 auto;
30    border-radius: 8px;
31  }
32
33  article h1,
34  article h2,
35  article h3 {
36    color: var(--article-heading);
37    margin-top: 24px;
38    margin-bottom: 12px;
39  }
40
41  article p {
42    line-height: 1.8;
43    margin-bottom: 16px;
44  }
45
46  article a {
47    color: var(--link-color);
48    text-decoration: none;
49    border-bottom: 1px dotted currentColor;
50  }
51
52  article code {
53    background: var(--code-bg);
54    color: var(--code-text);
```

```

55     padding: 2px 6px;
56     border-radius: 3px;
57     font-family: 'Courier New', monospace;
58 }
59
60 article pre {
61     background: var(--code-bg);
62     color: var(--code-text);
63     padding: 16px;
64     border-radius: 4px;
65     overflow-x: auto;
66     margin-bottom: 16px;
67 }
68
69 article blockquote {
70     border-left: 4px solid var(--blockquote-border);
71     padding-left: 20px;
72     margin-left: 0;
73     margin-bottom: 16px;
74     color: var(--article-text);
75     font-style: italic;
76 }
77
78 .post-meta {
79     color: var(--article-text);
80     opacity: 0.7;
81     margin-bottom: 24px;
82     font-size: 14px;
83 }
84 </style>
85
86 <article>
87     <h1>Understanding CSS Themes</h1>
88     <div class="post-meta">
89         Posted on January 15, 2024 by Author
90     </div>
91
92     <p>Dark mode has become an essential feature in modern web applications.
93         With <code>prefers-color-scheme</code>, implementing proper theme
94         support is straightforward.</p>
95
96     <h2>Getting Started</h2>
97     <p>The foundation of theme support starts with CSS custom properties:</p>
98
99     <pre><code>:root {
100 --bg: white;
101 --text: black;
102 }
103
104 @media (prefers-color-scheme: dark) {
105     :root {
106         --bg: #1a1a1a;
107         --text: white;
108     }

```



```
107 }</code></pre>
108
109     <blockquote>
110         Respecting user preferences isn't just nice to -haveit's essential for
111         accessibility and user experience.
112     </blockquote>
113
114     <p>Learn more about <a href="#">dark mode best practices</a>.</p>
115 </article>
```

Listing 4.2: Blog Post with Theme Support

Chapter 5

Advanced Techniques

5.1 Combining with Manual Theme Toggle

```
1 <style>
2   /* System preference defaults */
3   :root {
4     --bg: white;
5     --text: black;
6     color-scheme: light;
7   }
8
9   @media (prefers-color-scheme: dark) {
10    :root {
11      --bg: #1a1a1a;
12      --text: white;
13      color-scheme: dark;
14    }
15  }
16
17  /* Manual override (data attribute on html) */
18  html[data-theme="light"] {
19    --bg: white;
20    --text: black;
21    color-scheme: light;
22  }
23
24  html[data-theme="dark"] {
25    --bg: #1a1a1a;
26    --text: white;
27    color-scheme: dark;
28  }
29
30  /* Ignore system preference when override is set */
31  html[data-theme="light"] @media (prefers-color-scheme: dark) {
32    /* Doesn't apply - theme override takes precedence */
33  }
34
35  body {
36    background: var(--bg);
```

```

37     color: var(--text);
38   }
39 </style>
40
41 <script>
42   // Check for saved theme preference
43   const savedTheme = localStorage.getItem('theme');
44
45   if (savedTheme) {
46     // Use saved preference
47     document.documentElement.setAttribute('data-theme', savedTheme);
48   } else if (window.matchMedia('(prefers-color-scheme: dark)').matches) {
49     // Use system preference
50     document.documentElement.setAttribute('data-theme', 'dark');
51   } else {
52     document.documentElement.setAttribute('data-theme', 'light');
53   }
54
55   // Theme toggle button
56   document.getElementById('theme-toggle').addEventListener('click', () => {
57     const current = document.documentElement.getAttribute('data-theme') ||
58       'light';
59     const next = current === 'light' ? 'dark' : 'light';
60
61     document.documentElement.setAttribute('data-theme', next);
62     localStorage.setItem('theme', next);
63   });
64
65   // Listen for system preference changes
66   window.matchMedia('(prefers-color-scheme: dark)').addEventListener('change', (e) => {
67     if (!localStorage.getItem('theme')) {
68       // Only update if no manual override
69       const theme = e.matches ? 'dark' : 'light';
70       document.documentElement.setAttribute('data-theme', theme);
71     }
72   });
73 </script>
74 <button id="theme-toggle">Toggle Theme</button>

```

Listing 5.1: System Preference + Manual Override

5.2 Smooth Transitions

```

1 <style>
2   body {
3     background: var(--bg);
4     color: var(--text);
5     transition: background 0.3s ease, color 0.3s ease;
6   }
7

```

```
8     .card {
9         background: var(--bg-secondary);
10        border: 1px solid var(--border);
11        transition: background 0.3s ease, border-color 0.3s ease;
12    }
13
14    button {
15        background: var(--accent);
16        transition: background 0.2s ease;
17    }
18
19    /* Disable transitions during theme change via data attribute */
20    html.theme-switching,
21    html.theme-switching * {
22        transition: none !important;
23    }
24 </style>
25
26 <script>
27     function setTheme(theme) {
28         // Add class to disable transitions
29         document.documentElement.classList.add('theme-switching');
30
31         // Change theme
32         document.documentElement.setAttribute('data-theme', theme);
33
34         // Re-enable transitions
35         setTimeout(() => {
36             document.documentElement.classList.remove('theme-switching');
37         }, 50);
38     }
39 </script>
```

Listing 5.2: Smooth Color Transitions

5.3 Respecting prefers-reduced-motion

```
1 <style>
2     /* Always respect reduced motion */
3     @media (prefers-reduced-motion: reduce) {
4         * {
5             animation-duration: 0.01ms !important;
6             animation-iteration-count: 1 !important;
7             transition-duration: 0.01ms !important;
8         }
9     }
10
11     /* Only transition colors in non-reduced-motion preference */
12     @media (prefers-reduced-motion: no-preference) {
13         body {
14             transition: background 0.3s, color 0.3s;
15         }
16     }
```

```
16     }
17
18     /* Combine multiple preferences */
19     @media (prefers-color-scheme: dark) and (prefers-reduced-motion: no-
20         preference) {
21         body {
22             background: #1a1a1a;
23             color: white;
24             transition: all 0.3s ease;
25         }
26     }
27
28     @media (prefers-color-scheme: dark) and (prefers-reduced-motion: reduce) {
29         body {
30             background: #1a1a1a;
31             color: white;
32             /* No transitions */
33         }
34     }
35 </style>
```

Listing 5.3: Combining with Accessibility Preferences

Chapter 6

Images and Media

6.1 Responsive Images for Themes

```
1 <style>
2   .hero-image {
3     display: none;
4   }
5
6   @media (prefers-color-scheme: light) {
7     .hero-image.light {
8       display: block;
9     }
10  }
11
12  @media (prefers-color-scheme: dark) {
13    .hero-image.dark {
14      display: block;
15    }
16  }
17
18  /* Picture element approach (better) */
19  picture {
20    width: 100%;
21  }
22
23  picture img {
24    width: 100%;
25    display: block;
26  }
27 </style>
28
29 <!-- Simple approach -->
30 
31 
32
33 <!-- Better: Picture element -->
34 <picture>
35   <source srcset="hero-dark.png" media="(prefers-color-scheme: dark)" />
36   <source srcset="hero-light.png" media="(prefers-color-scheme: light)" />
```

```

37     
38 </picture>

```

Listing 6.1: Different Images Per Theme

6.2 SVG Color Adaptation

```

1  <style>
2      svg {
3          fill: var(--text);
4      }
5
6      svg.icon-accent {
7          fill: var(--accent);
8      }
9
10     svg.icon-error {
11         fill: var(--error);
12     }
13
14     svg.icon-success {
15         fill: var(--success);
16     }
17
18     :root {
19         --text: #000;
20         --accent: #0066cc;
21         --error: #d32f2f;
22         --success: #00a86b;
23     }
24
25     @media (prefers-color-scheme: dark) {
26         :root {
27             --text: #e0e0e0;
28             --accent: #4da6ff;
29             --error: #ff6b6b;
30             --success: #66cc99;
31         }
32     }
33 </style>
34
35 <!-- SVG that adapts to theme -->
36 <svg width="24" height="24" viewBox="0 0 24 24">
37     <circle cx="12" cy="12" r="10" />
38 </svg>
39
40 <!-- SVG with specific icon colors -->
41 <svg class="icon-accent" width="24" height="24" viewBox="0 0 24 24">
42     <path d="..." />
43 </svg>

```

Listing 6.2: SVG Icons with Theme Colors

Chapter 7

Real-World Examples

7.1 E-Commerce Product Page

```
1 <style>
2   :root {
3     --card-bg: white;
4     --card-border: #ddd;
5     --text: #333;
6     --text-light: #666;
7     --price: #d32f2f;
8     --button-bg: #0066cc;
9   }
10
11  @media (prefers-color-scheme: dark) {
12    :root {
13      --card-bg: #2d2d2d;
14      --card-border: #404040;
15      --text: #e0e0e0;
16      --text-light: #999;
17      --price: #ff6b6b;
18      --button-bg: #4da6ff;
19    }
20  }
21
22  .product-card {
23    background: var(--card-bg);
24    border: 1px solid var(--card-border);
25    border-radius: 8px;
26    padding: 20px;
27    max-width: 400px;
28  }
29
30  .product-image {
31    width: 100%;
32    background: var(--card-bg);
33    border-radius: 4px;
34    margin-bottom: 16px;
35    aspect-ratio: 1;
36  }
```



```
37
38 .product-title {
39     color: var(--text);
40     font-size: 20px;
41     font-weight: 600;
42     margin-bottom: 8px;
43 }
44
45 .product-description {
46     color: var(--text-light);
47     font-size: 14px;
48     margin-bottom: 16px;
49     line-height: 1.5;
50 }
51
52 .product-price {
53     color: var(--price);
54     font-size: 28px;
55     font-weight: bold;
56     margin-bottom: 16px;
57 }
58
59 .product-rating {
60     color: var(--text-light);
61     font-size: 14px;
62     margin-bottom: 16px;
63 }
64
65 .add-to-cart {
66     width: 100%;
67     background: var(--button-bg);
68     color: white;
69     border: none;
70     padding: 12px;
71     border-radius: 4px;
72     font-weight: 600;
73     cursor: pointer;
74     transition: opacity 0.2s;
75 }
76
77 .add-to-cart:hover {
78     opacity: 0.9;
79 }
80 </style>
81
82 <div class="product-card">
83     <div class="product-image"></div>
84     <h2 class="product-title">Premium Wireless Headphones</h2>
85     <p class="product-description">
86         High-quality audio with noise cancellation and 30-hour battery life.
87     </p>
88     <div class="product-price">\$199.99</div>
89     <div class="product-rating"> (1,234 reviews)</div>
90     <button class="add-to-cart">Add to Cart</button>
```

```
91 </div>
```

Listing 7.1: Product Page with Theme Support

7.2 Dashboard Interface

```
1 <style>
2   :root {
3     --dashboard-bg: #f5f5f5;
4     --panel-bg: white;
5     --panel-border: #e0e0e0;
6     --chart-line: #0066cc;
7     --text: #333;
8     --text-secondary: #666;
9   }
10
11   @media (prefers-color-scheme: dark) {
12     :root {
13       --dashboard-bg: #0f0f0f;
14       --panel-bg: #1e1e1e;
15       --panel-border: #333;
16       --chart-line: #4da6ff;
17       --text: #e0e0e0;
18       --text-secondary: #999;
19     }
20   }
21
22   .dashboard {
23     background: var(--dashboard-bg);
24     padding: 24px;
25   }
26
27   .dashboard-grid {
28     display: grid;
29     grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
30     gap: 20px;
31     margin-bottom: 20px;
32   }
33
34   .stat-card {
35     background: var(--panel-bg);
36     border: 1px solid var(--panel-border);
37     border-radius: 8px;
38     padding: 20px;
39   }
40
41   .stat-label {
42     color: var(--text-secondary);
43     font-size: 14px;
44     margin-bottom: 8px;
45   }
46
```

```

47     .stat-value {
48         color: var(--text);
49         font-size: 32px;
50         font-weight: bold;
51     }
52
53     .stat-change {
54         color: #00a86b;
55         font-size: 12px;
56         margin-top: 8px;
57     }
58
59     .chart-card {
60         background: var(--panel-bg);
61         border: 1px solid var(--panel-border);
62         border-radius: 8px;
63         padding: 20px;
64         height: 300px;
65     }
66
67     .chart-line {
68         stroke: var(--chart-line);
69     }
70 </style>
71
72 <div class="dashboard">
73     <h1>Dashboard</h1>
74
75     <div class="dashboard-grid">
76         <div class="stat-card">
77             <div class="stat-label">Total Revenue</div>
78             <div class="stat-value">\$45,230</div>
79             <div class="stat-change↑"> 12% from last month</div>
80         </div>
81
82         <div class="stat-card">
83             <div class="stat-label">Active Users</div>
84             <div class="stat-value">1,234</div>
85             <div class="stat-change↑"> 8% from last month</div>
86         </div>
87
88         <div class="stat-card">
89             <div class="stat-label">Conversion Rate</div>
90             <div class="stat-value">3.8%</div>
91             <div class="stat-change↑"> 0.3% from last month</div>
92         </div>
93     </div>
94
95     <div class="chart-card">
96         <h3>Revenue Trend</h3>
97         <svg class="chart-line" viewBox="0 0 400 300">
98             <!-- Chart SVG here -->
99         </svg>
100     </div>

```

```
101 </div>
```

Listing 7.2: Dashboard with Dark Mode Support

Chapter 8

Browser Compatibility and Detection

8.1 Feature Detection

```
1 <style>
2   /* Modern browsers support prefers-color-scheme */
3   @supports (display: flex) {
4     @media (prefers-color-scheme: dark) {
5       body {
6         background: #1a1a1a;
7         color: white;
8       }
9     }
10  }
11
12  /* Fallback for older browsers */
13  @supports not (display: flex) {
14    body {
15      background: white;
16      color: black;
17    }
18  }
19 </style>
20
21 <script>
22   // JavaScript feature detection
23   function supportsColorScheme() {
24     return window.matchMedia('(prefers-color-scheme: dark)').media !== '
25       not all';
26   }
27
28   if (supportsColorScheme()) {
29     console.log('prefers-color-scheme is supported');
30   } else {
31     console.log('prefers-color-scheme is not supported');
32     // Apply fallback
33   }
34
35   // Detect current preference
```

```
35     const prefersDark = window.matchMedia('(prefers-color-scheme: dark)').  
        matches;  
36     console.log('User prefers dark mode:', prefersDark);  
37 </script>
```

Listing 8.1: Detecting prefers-color-scheme Support

8.2 Browser Support Table

Browser	Version	Support	Notes
Chrome	76+	Full	Detects OS preference
Firefox	67+	Full	Detects OS preference
Safari	12.1+	Full	Detects OS preference
Edge	79+	Full	Detects OS preference
Opera	63+	Full	Detects OS preference
IE 11	All	No	Not supported

Chapter 9

Performance and Best Practices

9.1 Performance Tips

```
1 <style>
2   /* GOOD: Use CSS variables for theme */
3   :root {
4     --bg: white;
5     --text: black;
6   }
7
8   body { background: var(--bg); }
9
10  @media (prefers-color-scheme: dark) {
11    :root {
12      --bg: #1a1a1a;
13      --text: white;
14    }
15  }
16
17  /* AVOID: Duplicating selectors */
18  /* BAD: This causes unnecessary repaints */
19  body { background: white; color: black; }
20  @media (prefers-color-scheme: dark) {
21    body { background: #1a1a1a; color: white; }
22  }
23
24  /* GOOD: Group related properties */
25  /* Reduces selector specificity issues */
26  :root {
27    --palette-bg: white;
28    --palette-text: black;
29    --palette-border: #ddd;
30  }
31
32  @media (prefers-color-scheme: dark) {
33    :root {
34      --palette-bg: #1a1a1a;
35      --palette-text: white;
36      --palette-border: #333;
```

```
37     }  
38   }  
39 </style>
```

Listing 9.1: Performance Optimization

9.2 Best Practices Checklist

1. Use CSS custom properties for all theme values
2. Define defaults outside media queries
3. Ensure sufficient color contrast in both themes
4. Test with actual system dark mode, not just browser DevTools
5. Respect user accessibility preferences (prefers-reduced-motion)
6. Use `color-scheme` property in `meta` tag
7. Smooth transitions between themes (optional)
8. Avoid hardcoded colors in components
9. Test on multiple devices and browsers
10. Provide manual theme toggle if desired (optional)
11. Document color scheme in design system
12. Use semantic color names, not descriptive ones
13. Test images and media in both themes
14. Consider form elements and inputs
15. Update all interactive states (hover, focus, active)

Chapter 10

Advanced Color Strategies

10.1 Color-Mix Function (Modern)

```
1 <style>
2   :root {
3     --primary: #0066cc;
4     --primary-lighter: color-mix(in srgb, var(--primary) 20%, white);
5     --primary-darker: color-mix(in srgb, var(--primary) 80%, black);
6   }
7
8   @media (prefers-color-scheme: dark) {
9     :root {
10      --primary: #4da6ff;
11      --primary-lighter: color-mix(in srgb, var(--primary) 20%, white);
12      --primary-darker: color-mix(in srgb, var(--primary) 80%, black);
13    }
14  }
15
16  .button-primary {
17    background: var(--primary);
18  }
19
20  .button-primary:hover {
21    background: var(--primary-darker);
22  }
23
24  .button-ghost {
25    background: var(--primary-lighter);
26    color: var(--primary);
27  }
28 </style>
```

Listing 10.1: Color Mixing for Consistency

10.2 OKLCH Color Space

```
1 <style>
```

```
2      :root {
3          /* Light mode */
4          --surface: oklch(100% 0 0);
5          --on-surface: oklch(0% 0 0);
6          --primary: oklch(59% 0.2 259);
7          --accent: oklch(60% 0.19 56);
8      }
9
10     @media (prefers-color-scheme: dark) {
11         :root {
12             --surface: oklch(20% 0 0);
13             --on-surface: oklch(95% 0 0);
14             --primary: oklch(70% 0.2 259);
15             --accent: oklch(70% 0.19 56);
16         }
17     }
18
19     body {
20         background: var(--surface);
21         color: var(--on-surface);
22     }
23
24     .button {
25         background: var(--primary);
26         color: var(--on-surface);
27     }
28 </style>
```

Listing 10.2: Perceptually Uniform Colors

Chapter 11

Quick Reference

11.1 Complete Syntax Reference

```
1  /* Basic syntax */
2  @media (prefers-color-scheme: dark) { }
3  @media (prefers-color-scheme: light) { }
4  @media (prefers-color-scheme: no-preference) { }
5
6  /* Combined with other queries */
7  @media (prefers-color-scheme: dark) and (max-width: 768px) { }
8  @media (prefers-color-scheme: dark) or (prefers-reduced-motion: reduce) { }
9
10 /* Negation */
11 @media not (prefers-color-scheme: dark) { }
12
13 /* In media attribute */
14 <link rel="stylesheet" href="dark.css" media="(prefers-color-scheme: dark)">
15 <meta name="theme-color" content="#1a1a1a" media="(prefers-color-scheme: dark)">
16
17 /* JavaScript detection */
18 window.matchMedia('(prefers-color-scheme: dark)').matches
19 window.matchMedia('(prefers-color-scheme: dark)').addEventListener('change',
    fn)
20
21 /* color-scheme property */
22 html { color-scheme: light dark; }
23 html { color-scheme: dark; }
24
25 /* Form elements (auto-theming) */
26 input, textarea, select { color-scheme: light dark; }
```

Listing 11.1: prefers-color-scheme Syntax Cheat Sheet

11.2 Common Patterns

- Default light mode, override in dark media query

- Use CSS custom properties for all colors
- Combine with manual theme toggle
- Respect prefers-reduced-motion
- Use picture element for different images
- Apply color-scheme meta tag and CSS property
- Test on real devices, not just browser tools
- Use semantic color names in design system
- Ensure form elements are readable
- Consider graphs and charts visibility

11.3 CSS Properties Reference

```
1  /* color-scheme property - affects form elements */
2  body {
3      color-scheme: light dark;
4  }
5
6  /* Prefers light mode only */
7  body {
8      color-scheme: light;
9  }
10
11 /* Only dark mode */
12 body {
13     color-scheme: dark;
14 }
15
16 /* CSS custom properties */
17 :root {
18     --primary: #0066cc;
19 }
20
21 @media (prefers-color-scheme: dark) {
22     :root {
23         --primary: #4da6ff;
24     }
25 }
26
27 /* Use in properties */
28 body { color: var(--primary); }
```

Listing 11.2: Related Properties

Chapter 12

Conclusion

12.1 Summary

The `prefers-color-scheme` media query is an essential tool for modern web development, allowing websites to automatically adapt to user system preferences. Combined with CSS custom properties and good design practices, it enables beautiful, accessible experiences that respect user choices.

12.2 Key Takeaways

- Use `prefers-color-scheme` to detect OS dark mode
- Implement themes with CSS custom properties
- Define color palettes for light and dark modes
- Respect accessibility preferences (reduced motion)
- Use picture elements for theme-specific images
- Apply color-scheme property to form elements
- Test on actual devices, not just browser tools
- Consider combining with manual theme toggle
- Ensure accessible color contrast in all themes
- Keep CSS organized and maintainable

12.3 Related Features

Explore these complementary features:

- `prefers-reduced-motion` - Respect motion preferences
- `prefers-contrast` - High contrast mode detection
- `prefers-reduced-transparency` - Transparency preferences
- `color-scheme` - CSS property for semantic colors
- `@media (max-width)` - Responsive design queries

12.4 Resources

- [MDN Web Docs: prefers-color-scheme](#)
- [W3C Media Queries Level 5 Specification](#)
- [WCAG 2.1 Accessibility Guidelines](#)
- [CSS Custom Properties Specification](#)